MATHS MANTHAN

• TUITION CLASSES •

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Subject: Mathematics | **CBSE CLASS 10 – WEEKLY TEST 07 (SOLUTIONS)** | **Time:** 45 Minutes

Class: Class X (CBSE) | **Chapter 5: Arithmetic Progressions** | **MM: 20**

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CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, all intermediate algebraic steps, and clearly boxed final answers.
2. In internal choice (**OR**) questions, both options are solved completely with identical step-marking rigor.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 1.5M, 2M]**.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: Multiple Choice & Assertion-Reason MATHS MANTHAN | (5 × 1 = 5 Marks)

Q1. The next term of the AP $\sqrt{7}, \sqrt{28}, \sqrt{63}, \ldots$ is: [1]

(A) $\sqrt{70}$ (B) $\sqrt{84}$ (C) $\sqrt{98}$ (D) $\sqrt{112}$

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 1 Mark]

Correct Answer: **(D) $\sqrt{112}$** [0.5 Mark]

We simplify each radical term into simplest surd form and apply:

Formula / Theorem Used: $a_n = a + (n-1)d$

Justification: [0.5 Mark]

$\sqrt{7} = 1\sqrt{7}$, $\sqrt{28} = \sqrt{4\times 7} = 2\sqrt{7}$, $\sqrt{63} = \sqrt{9\times 7} = 3\sqrt{7}$. Here $a = \sqrt{7}$, $d = \sqrt{7}$.

Thus next (4th) term $= a_4 = 4\sqrt{7} = \sqrt{16\times 7} = \mathbf{\sqrt{112}}$.

Q2. Which term of the AP $121, 117, 113, \ldots$ is its first negative term? [1]

(A) 30th (B) 31st (C) 32nd (D) 33rd

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 1 Mark]

Correct Answer: **(C) 32nd term** [0.5 Mark]

Given: $a = 121$ and $d = 117 - 121 = -4$. For the first negative term:

Formula / Theorem Used: $a_n < 0 \iff a + (n-1)d < 0$

Justification: [0.5 Mark]

$121 + (n-1)(-4) < 0 \implies 125 - 4n < 0 \implies 4n > 125 \implies n > 31.25 \implies n = \mathbf{32}$.

Q3. If $x+1$, $3x$, and $4x+2$ are three consecutive terms of an AP, then the value of x is: [1]

(A) 1 (B) 2 (C) 3 (D) 4

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 1 Mark]

Correct Answer: **(C) 3** [0.5 Mark]

Using the property of three consecutive AP terms a, b, c:

Formula / Theorem Used: $2b = a + c$

Justification: [0.5 Mark]

$2(3x) = (x+1) + (4x+2) \implies 6x = 5x + 3 \implies 6x - 5x = 3 \implies x = \mathbf{3}$.

Q4. If the sum of the first n terms of an AP is $S_n = 2n^2 + 3n$, then its 5th term is: [1]

(A) 21 (B) 23 (C) 25 (D) 27

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 1 Mark]

Correct Answer: **(A) 21** [0.5 Mark]

Using the consecutive partial sums identity:

Formula / Theorem Used: $a_5 = S_5 - S_4$

Justification: [0.5 Mark]

$S_5 = 2(5)^2 + 3(5) = 50 + 15 = 65$, and $S_4 = 2(4)^2 + 3(4) = 32 + 12 = 44$.

Hence $a_5 = 65 - 44 = \mathbf{21}$.

Q5. **Assertion (A):** The common difference of an AP can be positive, negative, or zero.
Reason (R): If the common difference $d = 0$, all terms of the AP are identical real constants. [1]

(A) Both (A) and (R) are true, and (R) is correct explanation.

(B) Both (A) and (R) are true, and (R) is not correct explanation.

(C) Assertion (A) is true, but Reason (R) is false.

(D) Assertion (A) is false, but Reason (R) is true.

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 1 Mark]

Correct Answer: **(B) Both (A) and (R) are true; (R) is not correct explanation.** [0.5 Mark]

We examine the sign classification of common difference d:

Formula / Theorem Used: $d > 0$ (increasing AP), $d < 0$ (decreasing AP), $d = 0$ (constant AP)

Justification: [0.5 Mark]

Assertion (A) is true because $d \in \mathbb{R}$ may be positive, negative, or zero. Reason (R) is also true ($a_n = a + (n-1)(0) = a$), but (R) explains only the constant case ($d = 0$) and does not explain why d can be positive or negative. Hence option **(B)** holds.

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Q6. If the sum of the first n terms of an AP is $S_n = n(4n+1)$, find its nth term a_n and hence find its 10th term. **[2]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Given: $S_n = n(4n+1) = 4n^2 + n$.

Formula / Theorem Used: $a_n = S_n - S_{n-1} \qquad (n \geq 2)$

Step 1: Deriving General Term a_n [1 Mark]

Replacing n by $n-1$: $S_{n-1} = 4(n-1)^2 + (n-1) = 4(n^2 - 2n + 1) + n - 1 = 4n^2 - 8n + 4 + n - 1 = 4n^2 - 7n + 3$.

Subtracting:

$$a_n = S_n - S_{n-1} = (4n^2 + n) - (4n^2 - 7n + 3) = 4n^2 + n - 4n^2 + 7n - 3 = 8n - 3$$

(Verify at $n = 1$: $a_1 = 8(1) - 3 = 5$, and $S_1 = 1(4+1) = 5$, so $a_n = 8n - 3$ holds for all $n \geq 1$.)

Step 2: Finding the 10th Term [1 Mark]

Putting $n = 10$: $a_{10} = 8(10) - 3 = 80 - 3 = 77$.

$$\boxed{\mathbf{a_n = 8n - 3}, \qquad \mathbf{a_{10} = 77}}$$

Q7. If the sum of the first m terms of an AP is equal to the sum of its first n terms ($m \neq n$), prove that the sum of its first $(m+n)$ terms is zero. **[3]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Given: Let first term be a and common difference be d. Given $S_m = S_n$ with $m \neq n$.

We use the AP sum formula:

Formula / Theorem Used: $S_k = \frac{k}{2}[2a + (k-1)d]$

Step 1: Equating $S_m = S_n$ and Collecting Like Terms [1.5 Marks]

$$\frac{m}{2}[2a + (m-1)d] = \frac{n}{2}[2a + (n-1)d]$$

Multiplying both sides by 2 and expanding:

$$2am + m(m-1)d = 2an + n(n-1)d \implies 2a(m-n) + [(m^2 - n^2) - (m-n)]d = 0$$

Step 2: Factoring $(m-n)$ Using $m^2 - n^2 = (m-n)(m+n)$ [1 Mark]

$$(m-n)[2a + (m+n-1)d] = 0$$

Since $m \neq n$, we have $m - n \neq 0$. Dividing by $(m-n)$ gives $2a + (m+n-1)d = 0$. — (i)

Step 3: Evaluating S_{m+n} [0.5 Mark]

$$S_{m+n} = \frac{m+n}{2}[2a + (m+n-1)d] = \frac{m+n}{2} \times 0 = \boxed{\mathbf{0} \quad \text{[Hence Proved]}}$$

Q8. Solve the equation: $1 + 4 + 7 + 10 + \cdots + x = 287$. **[3]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Given: LHS is an AP $1, 4, 7, 10, \ldots, x$ with $a = 1$, $d = 4 - 1 = 3$, last term $a_n = x$, sum $S_n = 287$.

Formula / Theorem Used: $S_n = \frac{n}{2}[2a + (n-1)d]$ and $x = a_n = a + (n-1)d$

Step 1: Solving the Quadratic Equation for n [2 Marks]

$$\frac{n}{2}[2(1) + (n-1)(3)] = 287 \implies \frac{n}{2}[2 + 3n - 3] = 287 \implies n(3n-1) = 574$$

$$3n^2 - n - 574 = 0$$

Splitting the middle term ($3 \times (-574) = -1722 = (-42)(41)$ and $-42 + 41 = -1$):

$$3n^2 - 42n + 41n - 574 = 0 \implies 3n(n-14) + 41(n-14) = 0 \implies (n-14)(3n+41) = 0$$

Since n is a positive integer, reject $n = -\frac{41}{3}$; hence $n = 14$.

Step 2: Finding $x = a_{14}$ [1 Mark]

$$x = a_{14} = 1 + (14-1)(3) = 1 + 13 \times 3 = 1 + 39 = \boxed{\mathbf{40}}$$

Q9. If the mth term of an AP is $\frac{1}{n}$ and its nth term is $\frac{1}{m}$ $(m \neq n)$, show that the sum of its first (mn) terms is $\frac{1}{2}(mn + 1)$. **[3]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Given: $a_m = a + (m-1)d = \frac{1}{n}$ — (i) and $a_n = a + (n-1)d = \frac{1}{m}$ — (ii), with $m \neq n$.

Formula / Theorem Used: $a_k = a + (k-1)d$ and $S_k = \frac{k}{2}[2a + (k-1)d]$

Step 1: Solving for d and a [2 Marks]

Subtracting (ii) from (i):

$$(m-n)d = \frac{1}{n} - \frac{1}{m} = \frac{m-n}{mn} \implies d = \frac{1}{mn} \quad (\text{since } m \neq n)$$

Substituting $d = \frac{1}{mn}$ into (i):

$$a + \frac{m-1}{mn} = \frac{1}{n} \implies a = \frac{m}{mn} - \frac{m-1}{mn} = \frac{1}{mn}$$

Step 2: Computing S_{mn} [1 Mark]

$$S_{mn} = \frac{mn}{2}\left[2\left(\frac{1}{mn}\right) + (mn-1)\left(\frac{1}{mn}\right)\right] = \frac{mn}{2}\left[\frac{2 + mn - 1}{mn}\right] = \boxed{\frac{1}{2}(mn+1) \quad \text{[Hence Proved]}}$$

Q10. CASE STUDY: Handcrafted Circular Spiral in Textile Art

A textile artisan designs an ornamental spiral motif made up of successive semicircles, with centres alternately at A and B, starting with centre at A, of radii 0.5 cm, 1.0 cm, 1.5 cm, 2.0 cm, ... There are 13 consecutive semicircles in the complete motif. (Take $\pi = \frac{22}{7}$.) [4]

(i) State the formula for the length of a semicircle of radius r and find the length l_1 of the 1st semicircle. [1]

(ii) Find the radius and arc length l_{13} of the 13th semicircle. [1]

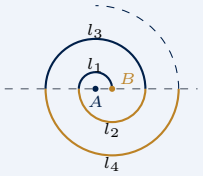
(iii) Compute the total length of the entire spiral made up of 13 consecutive semicircles. [2]

— **OR** —

Find the positive difference between the lengths of the 10th and 4th semicircles ($l_{10} - l_4$). [2]

▶ CASE-BASED SOLUTION & MARKING SCHEME: [Total: 4 Marks]

Extracted Scenario Data: Successive radii form the AP $0.5, 1.0, 1.5, \ldots, 6.5$ cm ($n = 13$). Each arc is a semicircle of length $l_k = \pi r_k$.



We apply the AP sum identity:

Formula / Theorem

Used: $l_k = \pi r_k$ and $S_{13} = \frac{13}{2}(l_1 + l_{13})$

Sub-part (i): Semicircle Length Formula and l_1 [1 Mark]

Length of a semicircle of radius r is $\mathbf{l = \pi r}$. With $r_1 = 0.5$ cm $= \frac{1}{2}$ cm:

$$l_1 = \frac{22}{7} \times \frac{1}{2} = \boxed{\mathbf{\frac{11}{7}}\text{ cm } (\text{or } \frac{\pi}{2}\text{ cm})}$$

Sub-part (ii): Radius and Length l_{13} of the 13th Semicircle [1 Mark]

$r_{13} = 0.5 + (13-1)(0.5) = 13 \times 0.5 = \mathbf{6.5}$ cm $= \frac{13}{2}$ cm, so:

$$l_{13} = \frac{22}{7} \times \frac{13}{2} = \boxed{\mathbf{\frac{143}{7}}\text{ cm } (\text{or } 20\frac{3}{7}\text{ cm})}$$

Sub-part (iii): Total Length of the 13-Semicircle Spiral [2 Marks]

$$S_{13} = l_1 + l_2 + \cdots + l_{13} = \pi(0.5 + 1.0 + 1.5 + \cdots + 6.5) = \pi \cdot \frac{13}{2}(0.5 + 6.5)$$

$$S_{13} = \frac{22}{7} \times \frac{13}{2} \times 7 = 11 \times 13 = \boxed{\mathbf{143}\text{ cm}}$$

Sub-part (iii) [OR Option]: Difference Between 10th and 4th Semicircle Lengths [2 Marks]

Here $r_{10} = 10 \times 0.5 = 5.0$ cm and $r_4 = 4 \times 0.5 = 2.0$ cm.

$$l_{10} - l_4 = \pi r_{10} - \pi r_4 = \pi(5.0 - 2.0) = \frac{22}{7} \times 3 = \boxed{\mathbf{\frac{66}{7}}\text{ cm } (\text{approx. } 9.43\text{ cm})}$$

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